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1 Data Report - TravelHunters Project

This document describes all data used in the TravelHunters project and ensures traceability and reproducibility of results.

The TravelHunters project collects and processes travel data from various sources to create a comprehensive dataset for hotels, activities, and destinations. Data is collected through web scraping, cleaned, and stored in a structured database.

1.1 Raw Data

1.1.1 Overview of Raw Datasets

Name	Source	Storage Location
Booking.com Hotels	Booking.com via Scrapy Spider	data_acquisition/json_final/booking_worldwide.json
Activities Data	GetYourGuide via Scrapy Spider	data_acquisition/json_final/activities_worldwide.json
Destinations Data	Tourism Websites via Scrapy Spider	data_acquisition/json_final/destinations_worldwide.json
Accommodations Data	Various Sources (Hostelworld, Airbnb)	data_acquisition/json_final/hotels.json
Google Maps Street View	Street View Images via Google API	onedrive
DuckDuckGo City Images	Images of Cities via DDG_search	onedrive
Wikipedia City Images	Wikipedia pages of cities	data_acquisition/json_final/wikipedia_images.json
GetYourGuide City Images	GetYourGuide Explorer Pages	data_acquisition/json_final/images.txt

1.1.1.1 Booking.com Hotels Dataset

- **Description:** Hotel data from Booking.com, including names, ratings, prices, locations, and image URLs.
- **Data Source:** Booking.com website via Scrapy Spider ([scraping_data_files/spiders/booking.py](#)).
- **Data Acquisition:** Automated web scraping with pagination for worldwide coverage.
- **Legal Aspects:** Publicly available data, respecting robots.txt.
- **Data Classification:** Public business data.
- **Variables:**
 - Hotel name
 - Rating
 - Price
 - Location
 - Image URLs
 - Description

1.1.1.2 Activities Dataset

- **Description:** Activities and tours from GetYourGuide, including prices, ratings, and locations.
- **Data Source:** GetYourGuide website via Scrapy([scraping_data_files/spiders/activities.py](#)).
- **Data Acquisition:** Web scraping with pagination for various destinations.
- **Data Classification:** Public business data.
- **Variables:**
 - Activity title
 - Price
 - Rating
 - Locationz
 - Image URLs
 - Source

1.1.2 Details of Image Datasets

1.1.2.1 GetYourGuide City Images

- **Description:** Images of cities and activities scraped from GetYourGuide Explorer pages.
- **Data Source:** GetYourGuide website via Selenium ([scraping_data_files/spiders/selescrape_img.py](#)).
- **Storage Location:** [data_acquisition/json_final/images.txt](#)
- **Variables:**
 - City name

- Image URL
- Activity link

1.1.2.2 DuckDuckGo City Images

- **Description:** Images of cities retrieved using DuckDuckGo search.
- **Data Source:** DuckDuckGo API ([scraping_data_files/spiders/cityImageScraper.py](#)).
- **Storage Location:** onedrive
- **Variables:**
 - City name
 - Image URL

1.1.2.3 Google Maps Street View Images

- **Description:** Street View images of hotels and destinations.
- **Data Source:** Google Maps API ([scraping_data_files/spiders/googlescraping.py](#)).
- **Storage Location:** onedrive
- **Variables:**
 - Latitude
 - Longitude
 - Image URL

1.1.2.4 Wikipedia City Images

- **Description:** Main images from Wikipedia pages of cities.
 - **Data Source:** Wikipedia pages via Scrapy Spider ([scraping_data_files/spiders/wikipedia_destinations.py](#)).
 - **Storage Location:** data_acquisition/json_final/wikipedia_images.json
 - **Variables:**
 - City name
 - Image URL
 - Description
 - Coordinates
-

1.2 Processed Data

1.2.1 Overview of Processed Datasets

Name	Source	Storage Location
Merged Travel Data	Consolidation of all raw data	data_acquisition/merged_travel_data.json
TravelHunters Database	Cleaned and structured data	database/travelhunters.db

1.2.2 Details of Processed Datasets

1.2.2.1 Merged Travel Data

- **Description:** Consolidated and cleaned data from all sources in a uniform format.
- **Processing Steps:**
 - Data loading from JSON files.
 - Validation and cleaning (URL validation, duplicate removal).
 - Data format normalization.
 - Consolidation into SQLite database.
- **Access:** Via data_cleaning_and_db_integration.py script.
- **Statistics:**
 - 2,000 hotels from Booking.com.
 - ~90 activities from GetYourGuide.
 - ~671 destinations.
 - Total: >2,700 records.
 - **Cities Used:** 129 unique cities.

1.3 Data Quality

1.3.1 Key Metrics

- **Completeness:** >95% of hotels have names and locations, ~80% have ratings.
 - **Consistency:** Uniform data formats through validation and cleaning.
 - **Accuracy:** Automatic validation of URLs and data formats.
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1.4 Entity Relationship Diagram

The ER diagram is located at `docs/er_diagramm/er_diagramm-ER.svg` and shows the relationships between Hotels, Activities, and Destinations.

1.5 Conclusion

This report provides a detailed overview of the data used in the TravelHunters project, including raw data sources, processed datasets, and data quality metrics. The structured and validated data serves as the foundation for building intelligent travel recommendation systems.